



Project Definition and Achievements

- Project Goal: To create realistic synthetic datasets for training Quality Assurance models through the use of image generation.
- Achievements:
 - Developed four-stage pipeline: Base Generation, Smart Labeling/Masking, Defect Injection, and Perspective Rectification.
 - Scale of Success: Generated a dataset of 7,200 images.
 - Successfully classified five distinct classes: (Good, Spiderweb, Scratch, Shattered Corner, and Puncture) with 99.03% accuracy.
- Novelty: comparative analysis between models trained on Raw vs. Rectified images

Puncture



Shattered corner



Scratch

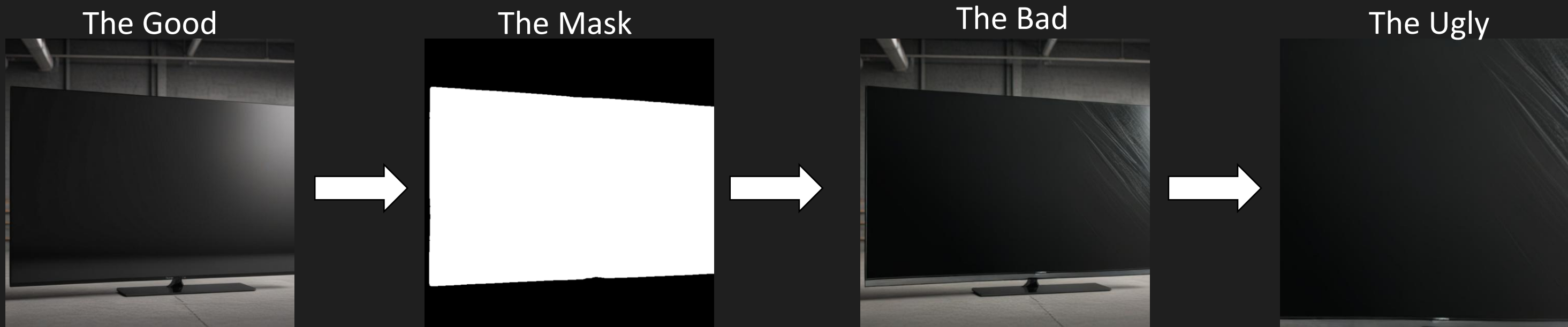


Spiderweb crack



Data Preparation and Quality Efforts

- Dataset Composition: We generated a total of 7,200 images: 3,600 Base Images and 3,600 Defected Images (900 for each of four defect categories).
- Synthetic Generation: SDXL Lightning was used to create base images of TVs in industrial environments.
- Precision Masking: Combined OWLv2 for object detection and SAM for segmentation.
- Perspective Rectification: A preprocessing script using OpenCV identified screen corners and applied perspective transformation for standardized, "flat" views.



- Fine-Tuning: Attempted to use Dreambooth to fine-tune a model on defect data, but the results did not meet quality standards.
- Localized Inpainting: Tested inpainting in small, specific areas, which created unnatural artifacts and "bad" visual transitions.
- Prompt Engineering: negative prompting ensured baseline screens were "clean" and suitable.

Dreambooth Failure



Without prompt engineering



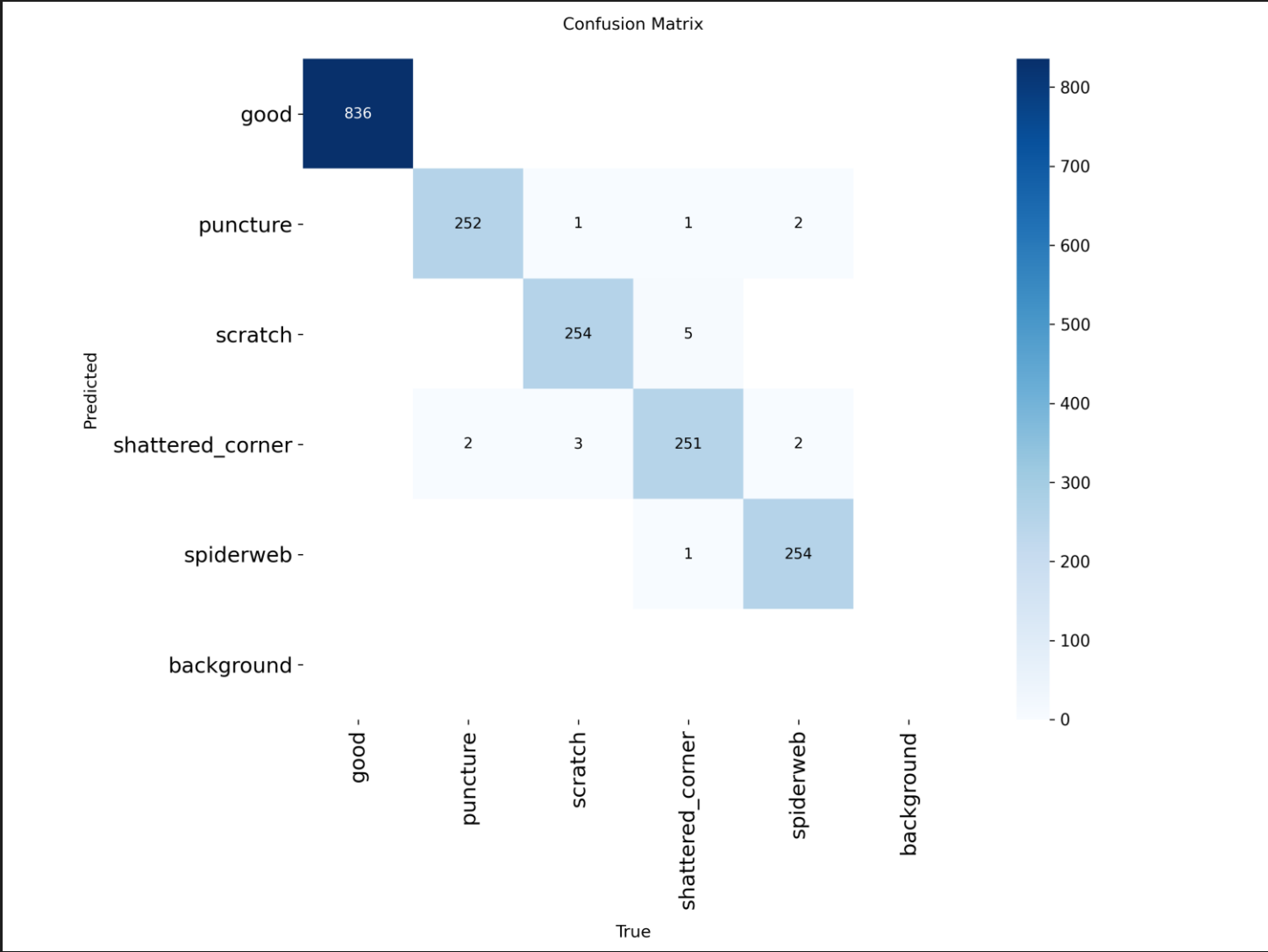
Methodology and Results (Comparative Analysis)

#The core of our review is the comparison between the Raw Dataset (with backgrounds) and the Rectified Dataset (isolated screens).

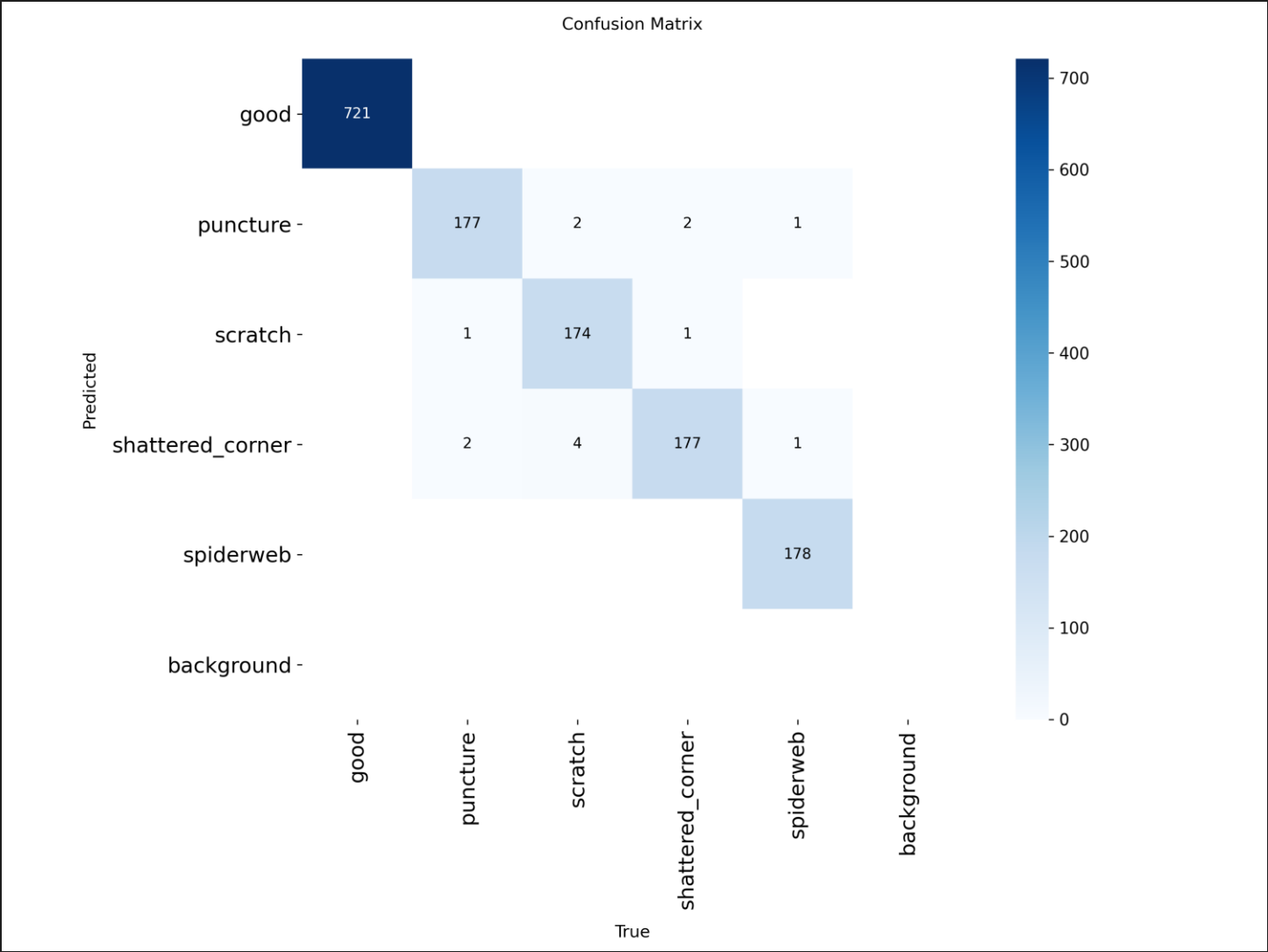
Metric	Raw Dataset	Rectified Dataset
Top-1 Accuracy	99.09%	99.03%
Training Time (10 Epochs)	2318.97 s	1308.96 s
Best Accuracy Epoch	10	7
Validation Loss	0.0293	0.0346

Key Insight: The Rectification step proved critical, reducing training time by 43% while maintaining comparable accuracy

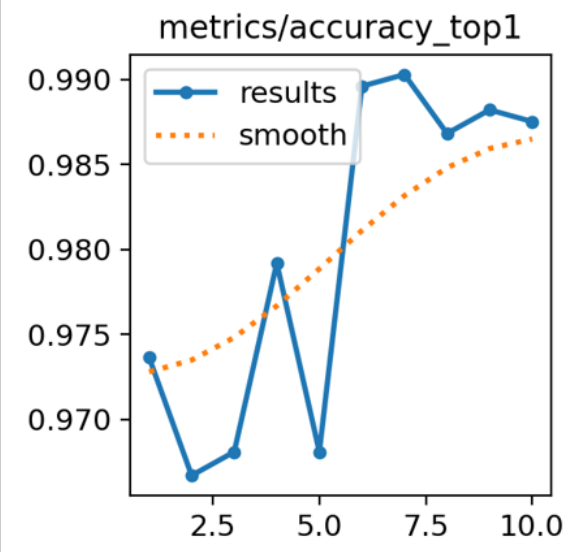
Raw pictures



Rectified pictures



Conclusions and Future Directions

Desired Results	Lessons Learned	Future Ideas																											
Achieved near-perfect accuracy 99.03%  <table border="1"><caption>metrics/accuracy_top1</caption><thead><tr><th>x-axis</th><th>results</th><th>smooth</th></tr></thead><tbody><tr><td>2.5</td><td>0.972</td><td>0.973</td></tr><tr><td>3.5</td><td>0.968</td><td>0.974</td></tr><tr><td>4.5</td><td>0.979</td><td>0.976</td></tr><tr><td>5.5</td><td>0.968</td><td>0.978</td></tr><tr><td>6.5</td><td>0.990</td><td>0.980</td></tr><tr><td>7.5</td><td>0.987</td><td>0.982</td></tr><tr><td>8.5</td><td>0.988</td><td>0.984</td></tr><tr><td>9.5</td><td>0.987</td><td>0.986</td></tr></tbody></table>	x-axis	results	smooth	2.5	0.972	0.973	3.5	0.968	0.974	4.5	0.979	0.976	5.5	0.968	0.978	6.5	0.990	0.980	7.5	0.987	0.982	8.5	0.988	0.984	9.5	0.987	0.986	Perspective transformation enables faster convergence Generative AI can effectively bridge the "Data Scarcity" gap for rare industrial defects	Synthetic-to-Real Validation 
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